Applied Physics -1

Scalars and Vectors - Lecture 02 (Module -01)

Scalars and Vectors

Scalar Quantity

A scalar is a physical quantity that has only **magnitude** and no direction.
Examples:

- Mass
- Time
- Temperature
- Energy
- Speed

Vector Quantity

A vector is a physical quantity that has both **magnitude and direction**.
Examples:

- Displacement
- Velocity
- Acceleration
- Force
- Momentum

Representation:

$$\vec{A} = A_x\hat{i} + A_y\hat{j} + A_z\hat{k}$$

Vector Arithmetic

Vector arithmetic refers to the mathematical operations performed on vectors. Since a vector has both magnitude and direction, vector operations must take both of these properties into consideration. The basic operations involving vectors are **vector addition, vector subtraction, and scalar multiplication**.
The most important point to remember is that vectors cannot be added or subtracted by considering their magnitudes alone. Their directions must also be taken into account.

Vector Addition

Vector addition is the process of combining two or more vectors to obtain a single vector called the **resultant vector**.
If two vectors $\vec{u}$ and $\vec{v}$ are added, the resultant vector $\vec{R}$ is written as

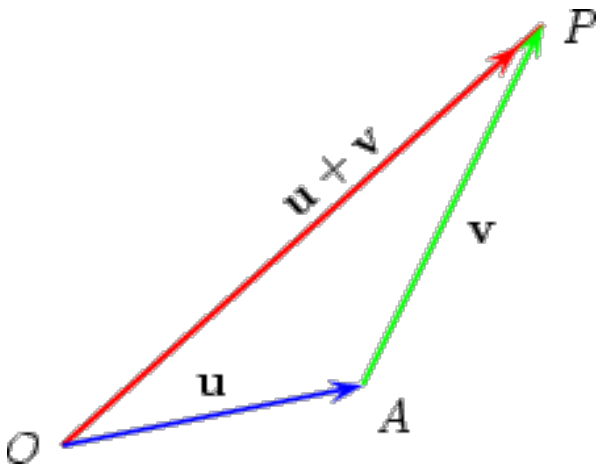
$$\boxed{\vec{R} = \vec{u} + \vec{v}}$$

The resultant represents the combined effect of the two vectors.
There are two commonly used graphical methods for adding vectors:

1. Triangle law of vector addition
2. Parallelogram law of vector addition

2.1.1 Triangle Law of Vector Addition

The triangle law states that if two vectors are represented by two sides of a triangle taken in order, then their resultant is represented by the third side of the triangle, taken from the initial point to the final point.
Consider the vectors $\vec{u}$ and $\vec{v}$ shown in the figure below.



In the figure, the vector $\vec{u}$ acts from point O to point A. The vector $\vec{v}$ then starts from point A and ends at point P. Therefore, the two vectors are arranged **head-to-tail**.
The resultant vector is the vector drawn from the starting point O to the final point P. Hence,

$$\boxed{\vec{OP} = \vec{OA} + \vec{AP}}$$

The diagonal or third side joining the initial point of the first vector to the final point of the second vector represents the resultant.
Therefore,

$$\boxed{\vec{R} = \vec{u} + \vec{v}}$$

The important idea is that the second vector is moved parallel to itself so that its tail coincides with the head of the first vector. Moving a vector in this manner does not change the vector because its magnitude and direction remain unchanged.

2.1.2 Understanding the Triangle Diagram

The diagram can be understood step by step as follows:

1. Start at point O and draw vector $\vec{u}$ towards point A.
2. From point A, draw vector $\vec{v}$ towards point P.
3. The vector directly connecting O to P is the resultant vector.
4. Therefore, the resultant is obtained by joining the tail of the first vector to the head of the second vector.

Thus,

$$\boxed{\vec{u} + \vec{v} = \vec{OP}}$$

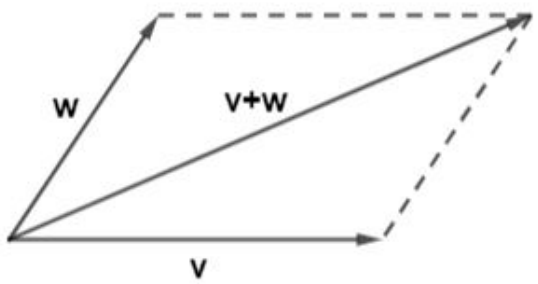
This method is particularly useful when several vectors are added successively.
For example, if

$$\vec{R} = \vec{u} + \vec{v} + \vec{w},$$

the vectors can be placed one after another in a head-to-tail arrangement. The resultant is then the vector joining the starting point to the final point.

2.1.3 Parallelogram Law of Vector Addition

The parallelogram law provides another graphical method for adding two vectors.
The law states that if two vectors starting at the same point are represented by two adjacent sides of a parallelogram, then their resultant is represented by the diagonal of the parallelogram passing through their common initial point.
Consider the vectors $\vec{v}$ and $\vec{w}$ shown in the figure.



In the figure, both vectors $\vec{v}$ and $\vec{w}$ start from the same initial point. A parallelogram is constructed by drawing lines parallel to each vector.
The diagonal of the parallelogram starting from the common initial point represents the resultant vector.
Therefore,

$$\boxed{\vec{R} = \vec{v} + \vec{w}}$$

The diagonal shown in the figure represents $\vec{v} + \vec{w}$.

2.1.4 Understanding the Parallelogram Diagram

The parallelogram method can be understood as follows:

1. Draw $\vec{v}$ and $\vec{w}$ from the same initial point.
2. Complete a parallelogram by drawing a line parallel to $\vec{v}$ through the head of $\vec{w}$.
3. Similarly, draw a line parallel to $\vec{w}$ through the head of $\vec{v}$.
4. The diagonal from the common initial point to the opposite corner gives the resultant.

Hence,

$$\boxed{\vec{R} = \vec{v} + \vec{w}}$$

The triangle and parallelogram methods give the same resultant. They are simply two different graphical ways of visualizing vector addition.

Vector Subtraction

Vector subtraction is performed by adding the negative of a vector.
For two vectors $\vec{A}$ and $\vec{B}$,

$$\boxed{\vec{R} = \vec{A} - \vec{B}}$$

can be written as

$$\boxed{\vec{R} = \vec{A} + (-\vec{B})}$$

Here, $-\vec{B}$ is the negative of $\vec{B}$.
The negative vector $-\vec{B}$ has:

- the same magnitude as $\vec{B}$, and
- the opposite direction to $\vec{B}$.

Therefore,

$$|\vec{B}| = |-\vec{B}|$$

but the directions are opposite.
For example,

$$\vec{A} - \vec{B} = \vec{A} + (-\vec{B}).$$

Thus, vector subtraction can be treated as vector addition after reversing the direction of the vector being subtracted.

2.2.1 Geometrical Interpretation of Vector Subtraction

Suppose we want to find

$$\vec{A} - \vec{B}.$$

The following steps are used:

1. Draw vector $\vec{A}$.
2. Reverse the direction of $\vec{B}$ to obtain $-\vec{B}$.
3. Add $\vec{A}$ and $-\vec{B}$ using the triangle or parallelogram method.
4. The resulting vector is $\vec{A} - \vec{B}$.

Therefore,

$$\boxed{\vec{A} - \vec{B} = \vec{A} + (-\vec{B})}$$

Scalar Multiplication

Scalar multiplication is the multiplication of a vector by a scalar quantity.
If a vector $\vec{A}$ is multiplied by a scalar k, then

$$\boxed{\vec{B} = k\vec{A}}$$

The scalar k changes the magnitude of the vector and, depending on its sign, may also change its direction.
The magnitude of the resulting vector is

$$\boxed{|\vec{B}| = |k||\vec{A}|}$$

The effect of scalar multiplication depends on the value of k.

2.3.1 Positive Scalar

If

$$k > 0,$$

the resulting vector has the same direction as $\vec{A}$.
For example, if k=2

$$\vec{B} = 2\vec{A}.$$

In this case,

$$|\vec{B}| = 2|\vec{A}|.$$

Therefore, the magnitude becomes twice the original magnitude while the direction remains unchanged.

2.3.2 Fractional Scalar

If

$$0 < k < 1,$$

the direction remains unchanged, but the magnitude decreases.
For example, if k=$1_{\overline{2}}$

$$\vec{B} = \frac{1}{2}\vec{A}.$$

Therefore,

$$|\vec{B}| = \frac{1}{2}|\vec{A}|.$$

The resulting vector has half the magnitude of the original vector and points in the same direction.

2.3.3 Negative Scalar

If

$$k < 0,$$

the direction of the resulting vector is reversed.
For example, if k=-2

$$\vec{B} = -2\vec{A}.$$

The magnitude is

$$|\vec{B}| = 2|\vec{A}|,$$

but the direction is opposite to that of $\vec{A}$.
Thus, a negative scalar both changes the magnitude and reverses the direction of the vector.

2.3.4 Zero Scalar

If

$$k = 0,$$

then

$$\boxed{0\vec{A} = \vec{0}}$$

where $\vec{0}$ is called the **zero vector**.
The zero vector has zero magnitude.

Vector Arithmetic Using Components

Vector arithmetic becomes particularly simple when vectors are expressed in terms of their Cartesian components. Consider two vectors

$$\vec{A} = A_x\hat{i} + A_y\hat{j} + A_z\hat{k}$$

and

$$\vec{B} = B_x\hat{i} + B_y\hat{j} + B_z\hat{k}.$$

2.4.1 Addition of Vectors

The corresponding components are added separately:

$$\begin{aligned}\vec{A} + \vec{B} = &(A_x + B_x)\hat{i} \\ &+ (A_y + B_y)\hat{j} \\ &+ (A_z + B_z)\hat{k}.\end{aligned}$$

Therefore,

$$\boxed{\vec{A} + \vec{B} = (A_x + B_x)\hat{i} + (A_y + B_y)\hat{j} + (A_z + B_z)\hat{k}}$$

2.4.2 Subtraction of Vectors

The corresponding components are subtracted:

$$\begin{aligned}\vec{A} - \vec{B} = &(A_x - B_x)\hat{i} \\ &+ (A_y - B_y)\hat{j} \\ &+ (A_z - B_z)\hat{k}.\end{aligned}$$

Therefore,

$$\boxed{\vec{A} - \vec{B} = (A_x - B_x)\hat{i} + (A_y - B_y)\hat{j} + (A_z - B_z)\hat{k}}$$

2.4.3 Scalar Multiplication Using Components

If a vector is multiplied by a scalar k, each component is multiplied by the same scalar:

$$\begin{aligned}k\vec{A} &= k(A_x\hat{i} + A_y\hat{j} + A_z\hat{k}) \\ &= (kA_x)\hat{i} + (kA_y)\hat{j} + (kA_z)\hat{k}.\end{aligned}$$

Thus,

$$\boxed{k\vec{A} = (kA_x)\hat{i} + (kA_y)\hat{j} + (kA_z)\hat{k}}$$

Important Properties of Vector Arithmetic

Vector arithmetic follows several important properties.

Commutative Property of Addition

The order of vector addition does not affect the resultant:

$$\boxed{\vec{A} + \vec{B} = \vec{B} + \vec{A}}$$

For example,

$$\vec{u} + \vec{v} = \vec{v} + \vec{u}.$$

This property can be understood from the parallelogram law because either vector can be taken as the first side of the parallelogram.

Associative Property of Addition

When three or more vectors are added, the grouping does not affect the result:

$$\boxed{(\vec{A}+\vec{B})+\vec{C}=\vec{A}+(\vec{B}+\vec{C})}$$

Additive Identity

Adding the zero vector to any vector gives the original vector:

$$\boxed{\vec{A}+\vec{0}=\vec{A}}$$

Additive Inverse

A vector added to its negative gives the zero vector:

$$\boxed{\vec{A}+(-\vec{A})=\vec{0}}$$

Vector Products

Vector products are mathematical operations involving two vectors. Unlike ordinary scalar multiplication, vector products must take into account both the magnitude and direction of the vectors.
There are two important types of vector products:

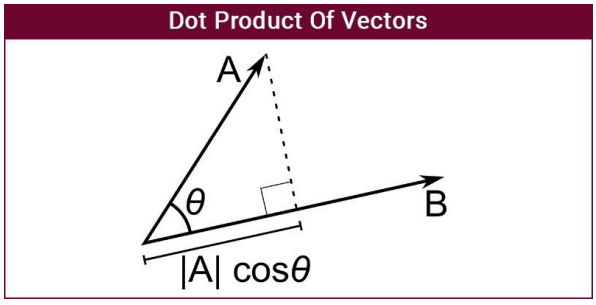
1. Dot product or scalar product
2. Cross product or vector product

The dot product produces a **scalar quantity**, whereas the cross product produces a **vector quantity**.

Dot Product (Scalar Product)

The **dot product** of two vectors is also called the **scalar product** because the result of the operation is a scalar quantity.
For two vectors $\vec{A}$ and $\vec{B}$ having magnitudes A and B, and making an angle θ with each other, the dot product is defined as

$$\boxed{\vec{A}\cdot\vec{B}=AB\cos\theta}$$



where

- $A=|\vec{A}|$ is the magnitude of $\vec{A}$,
- $B=|\vec{B}|$ is the magnitude of $\vec{B}$,
- θ is the angle between $\vec{A}$ and $\vec{B}$.

The result of the dot product is a scalar and therefore has no direction.

3.1.1 Sign of the Dot Product

The value of the dot product depends on the angle between the two vectors.

Case 1: $\theta=0^\circ$ When the two vectors are in the same direction,

$$\cos 0^\circ = 1.$$

Therefore,

$$\boxed{\vec{A}\cdot\vec{B}=AB}$$

The dot product is maximum and positive.

Case 2: $\theta=90^\circ$ When the two vectors are perpendicular,

$$\cos 90^\circ = 0.$$

Therefore,

$$\boxed{\vec{A}\cdot\vec{B}=0}$$

Thus, two non-zero perpendicular vectors have zero dot product.

Case 3: $\theta=180^\circ$ When the two vectors are in opposite directions,

$$\cos 180^\circ = -1.$$

Therefore,

$$\boxed{\vec{A}\cdot\vec{B}=-AB}$$

The dot product is negative.
Thus,

$$\boxed{\begin{array}{ll} 0<\theta<90^\circ & \Rightarrow \vec{A}\cdot\vec{B}>0 \\ \theta=90^\circ & \Rightarrow \vec{A}\cdot\vec{B}=0 \\ 180^\circ>\theta>90^\circ & \Rightarrow \vec{A}\cdot\vec{B}<0 \end{array}}$$

3.1.2 Dot Product in Component Form

If two vectors are written in Cartesian form as

$$\vec{A}=A_x\hat{i}+A_y\hat{j}+A_z\hat{k}$$

and

$$\vec{B}=B_x\hat{i}+B_y\hat{j}+B_z\hat{k},$$

then their dot product is

$$\boxed{\vec{A}\cdot\vec{B}=A_xB_x+A_yB_y+A_zB_z}$$

This is obtained using the properties of unit vectors:

$$\hat{i}\cdot\hat{i}=\hat{j}\cdot\hat{j}=\hat{k}\cdot\hat{k}=1$$

and

$$\hat{i}\cdot\hat{j}=\hat{j}\cdot\hat{k}=\hat{k}\cdot\hat{i}=0.$$

3.1.3 Application of Dot Product: Projection

The dot product can also be used to find the projection of one vector onto another.
The scalar projection of $\vec{A}$ along $\vec{B}$ is

$$\boxed{A_{\parallel} = A\cos\theta}$$

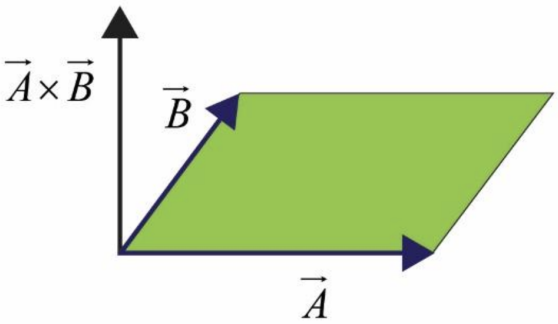
Alternatively, using the dot product,

$$\boxed{A_{\parallel} = \frac{\vec{A}\cdot\vec{B}}{|\vec{B}|}}$$

Thus, the dot product provides a convenient mathematical method for finding the component of one vector along another.

Cross Product (Vector Product)

The **cross product** of two vectors is also called the **vector product** because the result is a vector.
For two vectors $\vec{A}$ and $\vec{B}$ having magnitudes A and B, and making an angle θ, their cross product is defined as



$$\boxed{\vec{A}\times\vec{B} = AB\sin\theta\,\hat{n}}$$

where $\hat{n}$ is a unit vector perpendicular to both $\vec{A}$ and $\vec{B}$.
Unlike the dot product, the cross product produces a vector.

3.2.1 Direction of the Cross Product

The direction of the cross product is determined using the **right-hand rule**.
To determine the direction of

$$\vec{A}\times\vec{B},$$

follow these steps:

1. Point the fingers of your right hand in the direction of $\vec{A}$.
2. Curl your fingers towards $\vec{B}$ through the smaller angle θ.
3. Your thumb points in the direction of $\vec{A}\times\vec{B}$.

Thus,

$$\boxed{\vec{A}\times\vec{B}\perp\vec{A}}$$

and

$$\boxed{\vec{A}\times\vec{B}\perp\vec{B}}$$

The cross product therefore produces a vector perpendicular to both original vectors.

3.2.2 Magnitude of the Cross Product

The magnitude is

$$\boxed{|\vec{A}\times\vec{B}| = AB\sin\theta}$$

The value depends on the angle between the vectors.

Case 1: $\theta = 0^\circ$ For parallel vectors in the same direction,

$$\sin 0^\circ = 0.$$

Therefore,

$$\boxed{|\vec{A}\times\vec{B}| = 0}$$

Case 2: $\theta = 90^\circ$ For perpendicular vectors,

$$\sin 90^\circ = 1.$$

Therefore,

$$\boxed{|\vec{A}\times\vec{B}| = AB}$$

The magnitude of the cross product is maximum.

Case 3: $\theta = 180^\circ$ For vectors in opposite directions,

$$\sin 180^\circ = 0.$$

Therefore,

$$\boxed{|\vec{A}\times\vec{B}| = 0}$$

Thus, parallel or anti-parallel vectors have zero cross product.

3.2.3 Cross Product in Component Form

For

$$\vec{A} = A_x\hat{i} + A_y\hat{j} + A_z\hat{k}$$

and

$$\vec{B} = B_x\hat{i} + B_y\hat{j} + B_z\hat{k},$$

their cross product can be written using a determinant:

$$\boxed{\vec{A}\times\vec{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}}$$

Expanding the determinant,

$$\begin{aligned}\vec{A}\times\vec{B} = {} & (A_yB_z - A_zB_y)\hat{i} \\ & - (A_xB_z - A_zB_x)\hat{j} \\ & + (A_xB_y - A_yB_x)\hat{k}.\end{aligned}$$

Therefore,

$$\boxed{\vec{A}\times\vec{B} = (A_yB_z - A_zB_y)\hat{i} + (A_zB_x - A_xB_z)\hat{j} + (A_xB_y - A_yB_x)\hat{k}}$$

3.2.4 Application of Cross Product: Torque

One of the most important applications of the cross product is **torque**.
Torque is the turning effect produced by a force about a point or axis.
It is defined as

$$\boxed{\vec{\tau} = \vec{r} \times \vec{F}}$$

where

- $\vec{r}$ is the position vector from the axis of rotation to the point where the force acts,
- $\vec{F}$ is the applied force.

The magnitude of torque is

$$\boxed{\tau = rF\sin\theta}$$

The direction of torque is perpendicular to the plane containing $\vec{r}$ and $\vec{F}$ and is determined using the right-hand rule. The cross product is therefore appropriate for torque because torque has both magnitude and direction.

Properties of Cross Product

The cross product is **not commutative**. Reversing the order changes the direction:

$$\boxed{\vec{A} \times \vec{B} = -(\vec{B} \times \vec{A})}$$

It is distributive:

$$\boxed{\vec{A} \times (\vec{B} + \vec{C}) = \vec{A} \times \vec{B} + \vec{A} \times \vec{C}}$$

A vector crossed with itself is zero:

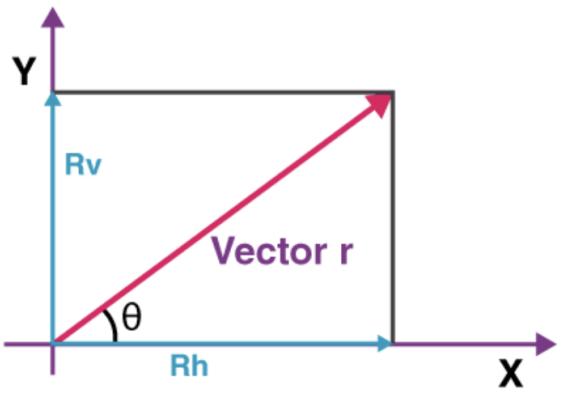
$$\boxed{\vec{A} \times \vec{A} = \vec{0}}$$

because

$$\sin 0^\circ = 0.$$

Vector Resolution

Vector resolution is the process of splitting a single vector into two or more component vectors along specified directions. The components together produce the original vector. The most commonly used form of vector resolution is the resolution of a vector into two mutually perpendicular components along the x- and y-axes.



Consider a vector $\vec{R}$ making an angle θ with the positive x-axis, as shown in the figure.
The vector $\vec{R}$ can be resolved into two perpendicular components:

$$\boxed{\vec{R} = R_x\hat{i} + R_y\hat{j}}$$

where

- R_x is the horizontal or x-component of $\vec{R}$,
- R_y is the vertical or y-component of $\vec{R}$.

Horizontal Component

The horizontal component is the projection of the vector $\vec{R}$ onto the x-axis.
From the right-angled triangle,

$$\cos\theta = \frac{\text{Adjacent side}}{\text{Hypotenuse}}.$$

Here,

$$\text{Adjacent side} = R_x$$

and

$$\text{Hypotenuse} = R.$$

Therefore,

$$\cos\theta = \frac{R_x}{R}.$$

Rearranging,

$$\boxed{R_x = R\cos\theta}$$

Thus, the horizontal component depends on both the magnitude of the original vector and the angle it makes with the horizontal axis.
As shown on the right side of the diagram, the horizontal component is labelled as

$$\boxed{R\cos(\theta)}$$

and acts along the positive x-direction when the vector lies in the first quadrant.

Vertical Component

The vertical component is the projection of the vector $\vec{R}$ onto the y-axis.
From the right-angled triangle,

$$\sin\theta = \frac{\text{Opposite side}}{\text{Hypotenuse}}.$$

Here,

$$\text{Opposite side} = R_y$$

and

$$\text{Hypotenuse} = R.$$

Therefore,

$$\sin\theta = \frac{R_y}{R}.$$

Rearranging,

$$\boxed{R_y = R\sin\theta}$$

As shown in the diagram, the vertical component is labelled

$$\boxed{R\sin(\theta)}$$

and acts along the positive y-direction when the vector lies in the first quadrant.

Relationship Between the Vector and Its Components

The original vector $\vec{R}$ is obtained by combining its horizontal and vertical components.
Therefore,

$$\boxed{\vec{R} = R_x\hat{i} + R_y\hat{j}}$$

Substituting the component expressions,

$$\boxed{\vec{R} = R\cos\theta\,\hat{i} + R\sin\theta\,\hat{j}}$$

This equation shows that a vector can be completely represented using its two perpendicular components.

Magnitude of the Resultant Vector

Since R_x and R_y are perpendicular components, the magnitude of the original vector can be obtained using the Pythagorean theorem.

$$R^2 = R_x^2 + R_y^2$$

Therefore,

$$\boxed{R = \sqrt{R_x^2 + R_y^2}}$$

This equation is useful when the horizontal and vertical components of a vector are known and the magnitude of the original vector needs to be found.

Direction of the Vector

The direction of the vector can also be determined from its components.
From the right-angled triangle,

$$\tan\theta = \frac{\text{Opposite side}}{\text{Adjacent side}}.$$

Therefore,

$$\tan\theta = \frac{R_y}{R_x}.$$

Hence,

$$\boxed{\theta = \tan^{-1}\left(\frac{R_y}{R_x}\right)}$$

Thus, if the horizontal and vertical components are known, both the magnitude and direction of the original vector can be calculated.

Sign Convention for Vector Components

The signs of the components depend on the quadrant in which the vector lies.
For a standard Cartesian coordinate system:

Quadrant	R_x	R_y
I	Positive	Positive
II	Negative	Positive
III	Negative	Negative
IV	Positive	Negative

For example, a vector in the first quadrant has

$$R_x > 0$$

and

$$R_y > 0.$$

For a vector in the second quadrant,

$$R_x < 0$$

while

$$R_y > 0.$$

Therefore, while resolving a vector, both the magnitude and the direction of each component must be considered.

Application: Robotics

Vector resolution is also important in robotics.
A robotic arm may exert a force in an arbitrary direction. This force can be resolved into horizontal and vertical components:

$$\vec{F} = F_x\hat{i} + F_y\hat{j}.$$

The individual components can then be used to determine the required motion, force, or torque at different joints.
Similarly, the velocity and acceleration of a robotic end-effector can be resolved into independent components for motion planning and control.

Differentiation

Differentiation is a mathematical process used to determine the **rate at which one quantity changes with respect to another**. It is one of the fundamental concepts of calculus and has wide applications in physics, engineering, mathematics, and computer science.
If a quantity y depends on another quantity x, then the rate of change of y with respect to x is represented by

$$\boxed{\frac{dy}{dx}}$$

Here, dy represents a small change in y, while dx represents a small change in x.
Thus,

$$\frac{dy}{dx} = \frac{\text{change in } y}{\text{change in } x}$$

in the limiting sense.

Important Differentiation Formulas

The following table summarizes the most commonly used differentiation formulas.

Function	Derivative
C	0
x^n	nx^{n-1}
cx^n	cnx^{n-1}
x	1
$\sqrt{x}$	$\dfrac{1}{2\sqrt{x}}$
$\dfrac{1}{x}$	$-\dfrac{1}{x^2}$
e^x	e^x
e^{ax}	ae^{ax}
$\ln x$	$\dfrac{1}{x}$
$\log_a x$	$\dfrac{1}{x \ln a}$
$\sin x$	$\cos x$
$\cos x$	$-\sin x$
$\tan x$	$\sec^2 x$
$\cot x$	$-\operatorname{cosec}^2 x$
$\sec x$	$\sec x \tan x$
$\operatorname{cosec} x$	$-\operatorname{cosec} x \cot x$

Important Differentiation Rules

The following rules are useful when differentiating more complicated functions.

Rule	Formula
Constant Rule	$\dfrac{d}{dx}(C) = 0$
Power Rule	$\dfrac{d}{dx}(x^n) = nx^{n-1}$
Constant Multiple Rule	$\dfrac{d}{dx}[cf(x)] = c\dfrac{df}{dx}$
Sum Rule	$\dfrac{d}{dx}[f+g] = f' + g'$
Difference Rule	$\dfrac{d}{dx}[f-g] = f' - g'$
Product Rule	$\dfrac{d}{dx}(uv) = u'v + uv'$
Quotient Rule	$\dfrac{d}{dx}\left(\dfrac{u}{v}\right) = \dfrac{vu' - uv'}{v^2}$
Chain Rule	$\dfrac{d}{dx}f(g(x)) = f'(g(x))g'(x)$

Integration

Integration is a mathematical process used to determine the **total accumulation of a quantity**. It is closely related to differentiation and can be understood as the **reverse process of differentiation**.
If differentiation determines the rate of change of a quantity, integration can be used to determine the original quantity from its rate of change.
The general form of integration is

$$\boxed{\int f(x)\,dx}$$

where $f(x)$ is the function being integrated and dx indicates that the integration is performed with respect to x.

Important Integration Formulas

The following table summarizes the most commonly used integration formulas.

Function	Integral
C	$Cx + C$
x^n	$\dfrac{x^{n+1}}{n+1} + C, \quad n \neq -1$
$\dfrac{1}{x}$	$\ln \lvert x \rvert + C$
e^x	$e^x + C$
e^{ax}	$\dfrac{1}{a}e^{ax} + C$
$\sin x$	$-\cos x + C$
$\cos x$	$\sin x + C$
$\sec^2 x$	$\tan x + C$
$\operatorname{cosec}^2 x$	$-\cot x + C$
$\sec x \tan x$	$\sec x + C$
$\operatorname{cosec} x \cot x$	$-\operatorname{cosec} x + C$

Important Integration Rules

Rule	Formula
Constant Multiple Rule	$\int cf(x)\,dx = c\int f(x)\,dx$
Sum Rule	$\int [f(x) + g(x)]\,dx = \int f(x)\,dx + \int g(x)\,dx$
Difference Rule	$\int [f(x) - g(x)]\,dx = \int f(x)\,dx - \int g(x)\,dx$
Substitution	$u = g(x), \quad du = g'(x)\,dx$
Integration by Parts	$\int u\,dv = uv - \int v\,du$

Worksheet Questions

1. A particle moves along a straight line according to the position function

$$x(t) = 4t^3 - 3t^2 + 6t + 2,$$

where x is measured in metres and t in seconds.
Find:

(a) the velocity of the particle as a function of time,
(b) the acceleration of the particle,
(c) the velocity and acceleration at $t = 2\,\text{s}$.

2. Evaluate the following indefinite integral:

$$\int \left(6x^4 - 8x^3 + 5x^2 - 4x + 7\right)\, dx.$$

3. The velocity of a particle is given by

$$v(t) = 6t^2 - 4t + 3 \quad \text{m/s}.$$

If the particle starts from the position $x = 2\,\text{m}$ at $t = 0$, determine its position at $t = 3\,\text{s}$.

4. Differentiate the following function with respect to x:

$$y = 3x^4 - 5x^2 + 7\sin x + 4e^x - 6\ln x.$$

Hence, find the value of $\dfrac{dy}{dx}$ at $x = 1$.

5. The displacement of a particle is given by

$$x(t) = t^4 - 4t^3 + 6t^2 + 2t.$$

Determine:

(a) the velocity $v(t)$,
(b) the acceleration $a(t)$,
(c) the times at which the particle is momentarily at rest,
(d) the acceleration at each of those instants.

6. Evaluate the integral

$$\int x(3x^2 + 2)^4 \, dx$$

using an appropriate integration method.

7. A force acting on a particle varies with position according to

$$F(x) = 4x^3 - 6x + 8 \quad \text{N}.$$

Calculate the work done by this force in moving the particle from

$$x = 1\,\text{m}$$

to

$$x = 3\,\text{m}.$$

Use

$$W = \int_{x_1}^{x_2} F(x) \, dx.$$

8. The velocity of a vehicle is described by

$$v(t) = 10e^{-0.2t} + 2 \quad \text{m/s}.$$

Determine the total displacement of the vehicle during the first 10 s.

9. A particle has acceleration

$$a(t) = 12t^2 - 6t + 4 \quad \text{m/s}^2.$$

Its initial velocity is $v(0) = 5\,\text{m/s}$ and its initial position is $x(0) = 2\,\text{m}$.
Using integration, determine:

(a) the velocity function $v(t)$,
(b) the position function $x(t)$,
(c) the position of the particle at $t = 2\,\text{s}$.

10. A rectangular sheet has a fixed perimeter of 40 cm. If its length is x cm, express its area A as a function of x.
Using differentiation:

(a) determine the value of x for which the area is maximum,
(b) determine the corresponding width,
(c) calculate the maximum possible area.

11. In robotics, the position of a robot's end-effector can be represented as a time-dependent function. Consider a simplified one-dimensional model:

$$x(t) = 2t^3 - 9t^2 + 12t.$$

Investigate how differentiation can be used to determine the velocity and acceleration of the end-effector.

Using the given function:

(a) derive $v(t)$ and $a(t)$,
(b) determine the times at which the end-effector is stationary,
(c) explain how velocity and acceleration information can be useful in robotic motion planning and control.

Support your explanation with at least one real-world robotics example.

12. Variable forces are encountered in engineering systems such as springs, robotic actuators, and mechanical mechanisms. Consider a force that varies with displacement according to

$$F(x) = kx,$$

where k is a constant.

Using integration:

(a) derive an expression for the work done in moving an object from $x = 0$ to $x = X$,
(b) explain why integration is required when the force is not constant,
(c) investigate one engineering application where the work done by a variable force is calculated using integration.

For the numerical case

$$k = 500\,\mathrm{N/m}, \qquad X = 0.20\,\mathrm{m},$$

calculate the work done.